

Apply to be a Science Corps Fellow - Entry #3335

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When did you or do you expect to complete your PhD?
August 2025
Please tell us briefly about your university education (institution, city, country, year)
I am a PhD candidate in Aerospace Engineering at the University of Illinois at Urbana-Champaign in the United States, where I also earned my BS in Aerospace Engineering in 2020. Before that, I completed an Associate's Degree in Engineering Science at the College of Lake County in Grayslake, Illinois in 2017. My PhD research focuses on the response of ablative materials in high-temperature, chemically reactive hypersonic flows during atmospheric entry. I designed and built a lab-scale flow reactor combining a capacitively coupled, radio-frequency plasma source with resistively-heated graphite samples to study carbon oxidation kinetics, improving heat shield predictions for future spacecraft. Beyond my PhD, I earned a private pilot's license for fixed wing, single engine aircraft in 2022 at the Parkland Institute of Aviation and I am currently training to become a certified Emergency Medical Technician at Parkland College in Champaign, Illinois.
Why would you like to be a Science Corps Fellow?
The Science Corps Fellowship aligns with two of my deepest passions: cross-cultural exchange and the empowerment of people that may not see science as a viable path for themselves. Attending a world-renowned university and scientific research institution so close to where I grew up has been a privilege, and I am deeply thankful for the opportunity to collaborate with so many brilliant people from all over the world. Through these interactions, I have gained exposure to a wide variety of traditions, customs, and perspectives that otherwise would have remained foreign and mysterious. I have also admired the courage of my international colleagues who have left behind the comforting familiarity of home to work and study in another country. I often wondered if I would have the same resilience to fully immerse myself in another culture and now, I am ready to take the next step. I want to experience an entirely different educational system and understand the challenges that students face in other

communities. By stepping outside of my familiar environment, I will become a more well-rounded scientist and communicator.

Growing up, I was fortunate to receive unwavering encouragement from my parents, teachers, and peers to confidently pursue a career in science and engineering. However, I have since realized that this was not the case for all people or even for all the members of my own community. I find this deeply troubling because scientific curiosity is a fundamental part of human nature, yet many people are deprived of the chance to explore it simply due to the circumstances of their birth. I firmly believe that science thrives on diversity of thought, and our collective understanding of the world is diminished when entire demographics are excluded from the scientific process.

As a Science Corps Fellow, I see an irresistible opportunity to help bridge this gap. I am eager to share my passion for discovery with young minds who may have never imagined themselves as future scientists, while also immersing myself in a new culture and learning from the community I serve. By engaging with students in scientifically undeserved areas, I hope to inspire the same curiosity and confidence that was instilled in me, showing them that science is for everyone!

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

I recognize that not everyone has had the same access to science that I did. I was fortunate to have supportive parents and passionate teachers who nurtured my curiosity and encouraged me to pursue STEM. Their guidance was instrumental in shaping my path, and I would not be where I am today without their support. However, through my higher education, I have become increasingly aware of the disparities in access to science and engineering programs between people who look like me and who come from backgrounds like mine and those with very different life experiences.

In response, I have sought ways to help correct these inequities, however small my contribution may be. Within the Aerospace Engineering department at the University of Illinois, I have worked to foster a more inclusive community and advocate for underrepresented groups. I have been actively involved with Minorities in Aerospace (MAero) and Women in Aerospace (WIA), student organizations dedicated to promoting diversity and supporting marginalized voices within our field. My primary contribution has been mentoring younger students by sharing my experiences, offering academic guidance, and helping them navigate the path to internships and careers in STEM.

One experience that has especially prepared me for the Science Corps Fellowship is my work with WIA on the Girls Learning Air and Space Science (GLASS) program. This event engages young girls (ages 8–16) in hands-on activities that teach the fundamentals of aerodynamics and spacecraft design. As a supporting volunteer, I help prepare supplies and assist students as they explore key aerospace concepts. This involves taking complex scientific ideas and breaking them down into intuitive, fundamental concepts. Seeing their enthusiasm for air and space travel is incredibly rewarding, reinforcing my belief in the power of early exposure to science. Their excitement reignited my own curiosity and sense of wonder, reminding me why I chose this field in the first place.

I believe the work of a Science Corps Fellow will be even more impactful, providing me with the opportunity to bring that same sense of excitement and possibility to students who may never have considered a future in science. My knowledge and experience conducting research experiments in a lab has taught me the importance of engaging

students with interactive learning to make science tangible and exciting, which is absolutely an approach I would bring to the Science Corps. Through mentorship, hands-on learning, and cultural exchange, I hope to inspire the next generation of scientists and engineers, just as my mentors once inspired me.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia

In science education

Other

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

My ultimate career aspiration is to become an astronaut. To me, spaceflight represents the greatest adventure and the culmination of everything I have worked for, both professionally and personally. I know my chances are low. Very few opportunities exist to fly in space and there are many brilliant, talented, and competitive people who are just as eager to go as I am. This fact doesn't discourage me. I believe that a life lived in pursuit of such an ambitious goal will propel me toward achievements I may never have even considered. When I retire, I will look back with pride on a career filled with advances in space exploration, even if I never became an astronaut.

I have already applied for the NASA astronaut program three times over the past eight years and, despite rejections, I will continue applying each time the selection opens. In the near term (~2 years after the completion of my PhD degree this year) I expect to begin a postdoctoral program to further develop my research expertise. While my current work on ablative materials for atmospheric entry is highly engaging, I would like to expand my scientific background and explore other aspects of space flight, particularly the physiological effects of prolonged space travel. Regardless of the specific research area, I know that I want to remain in a hands-on laboratory setting where I can operate and maintain experimental equipment.

In addition to my aerospace research, I am currently learning basic life support skills to become certified as an Emergency Medical Technician. While this may appear to be a departure from my scientific career, I view it as an exciting and complementary skill set that aligns with my aspirations for spaceflight. Medical emergencies can occur anywhere, even in space, and having the ability to provide lifesaving care is invaluable. This work could be applied in my career in a variety of ways including a part-time job, volunteering with a local fire department, or during an emergency situation in any environment. This is an important step to achieving my long-term goals.

The opportunity to teach as a Science Corps Fellow would allow me to improve another set of skills which are an important responsibility of an astronaut: to be a compelling scientific communicator and educator. Teaching under a new cultural context would challenge me to adapt my communication style, making me a stronger educator and global advocate for science. Looking further ahead (~10 years), I aim to contribute to human spaceflight efforts while working with NASA, a private space company, or an international program. It would be extraordinarily satisfying to support a real human spaceflight mission, whether it be in low-Earth orbit, on the Moon, or to Mars. No matter what role I take on in the future – scientist, engineer, or astronaut – I am committed to supporting humanity's exploration into space.

Please upload your CV (.pdf)



[NAAnderson-CV.pdf](#)

How did you find out about Science Corps?

An email list in your institution

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